QF620 Stochastic Modelling in Finance — Session 3 Reference Cheatsheet

Brownian Motion and Martingale: random walks, the scaled-walk limit, the definition and properties of Brownian motion, the Markov property, geometric Brownian motion, Brownian martingales

Built from Session 3 slides (Tee Chyng Wen, SMU) · October 2026 · revised edition with intuition and full slide derivations

Contents

Notation used throughout (slide conventions) **2**

0. Topic map **2**

1 Concept entries **3**

1.1 Symmetric random walk [S2] . . . 3
1.2 History: from pollen to stock prices [S3–S8] . . . 3
1.3 From random walk to Brownian motion [S9–S10] . . . 3
1.4 Mathematical definition of Brownian motion [S11] . . . 4
1.5 Example: $\sqrt{t}\,Z$ is not a Brownian motion [S12–S13] . . . 4
1.6 Statistical properties and joint normality [S14] . . . 5
1.7 Probability examples [S15] . . . 5
1.8 Markov property [S16] . . . 6
1.9 Geometric Brownian motion [S17] . . . 6
1.10 Path properties [S18–S20] . . . 7
1.11 Brownian martingales [S21–S23] . . . 7
1.12 Exponential martingale [S24–S25] . . . 8
1.13 Implementation *[Beyond slides]* (tested) . . . 9

2 Comparison tables **9**

2.1 Brownian motion vs $\sqrt{t}\,Z$ [S12–S13] . . . 9
2.2 Random walk vs Brownian motion . . . 9
2.3 Martingale or not? . . . 9
2.4 Path properties . . . 9

3 Master formula sheet **9**

4 Glossary **10**

5 Self-test **10**

Questions . . . 10
Solutions . . . 10

Appendix A. Slide coverage checklist **11**

Appendix B. Flag summary **11**

Source: "Session 3: Brownian Motion and Martingale" (Tee Chyng Wen, SMU, 25 slides).

How to read this sheet. Each concept entry has up to five parts: **The idea in words** (what the concept is and why it is there), the formula with a shaded where box naming every symbol, **Derivation** (the slide's

argument line by line, with the reason for each step in brown), **Worked example**, and **Exam angle** (traps and what to be able to reproduce). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

What you must be able to do after this session: state the four defining properties of Brownian motion and test a candidate process against them; compute mean, variance and covariance of W_t; do probability calculations by splitting into independent increments; show the Markov property; compute $\mathbb{E}[e^{\sigma W_t}]$ via the MGF; and prove or disprove the martingale property with the "add and subtract" trick $W_t = (W_t - W_s) + W_s$.

Notation used throughout (slide conventions)

Symbol	Meaning	Notes
S_n	position of a symmetric random walk after n steps	[S2]
X_i	i.i.d. ± 1 steps, probability $\frac{1}{2}$ each	[S9]
$W_n(t)$	scaled random walk with n steps on $[0,t]$	[S9]
W_t	(standard) Brownian motion / Wiener process at time t	[S11]
s, t, T	times, usually $0 \le s \le t \le T$	
$W_t - W_s$	increment of W over $[s,t]$	$\sim \mathcal{N}(0, t-s)$ [S11]
$\mathbb{E}_t[\cdot]$	expectation conditional on information up to time t (the path up to t)	[S16]
M_t	martingale	[S21]
$X_t = e^{W_t}$	geometric Brownian motion (slide's simplest form)	[S17]
σ	real constant (volatility)	[S24]

Notation trap: in Session 2 S_n was a stock price and X_i a coin toss; here S_n is the random-walk position on [S2] and X_t a generic process. $\mathbb{E}_s[\cdot]$ means $\mathbb{E}[\,\cdot \mid \mathcal{F}_s]$.

0. Topic map

```
PREREQUISITES: S1 normal MGF, lognormal mean ; S2 arithmetic tree σ√±(T/n) + CLT, conditional expectation rules
    ▼
SESSION 3├─
 BROWNIAN [–S2S10]│
     ├─ symmetric random walk: 2^n paths, range –[n, n]                    [S2]│
     ├─ history: Brown (pollen), Einstein, Bachelier 1900 (stock prices)   [–S3S8]│
     └─ scaled walk W_n: steps √±(t/n) every t/n ; CLT ⇒ W_n(t) → N(0,t)    [–S9S10]├─
 DEFINITION OF W_t [–S11S15]│
     ├─ W_0=0 ; continuous ; stationary independent increments ; N(0,–ts)  [S11]│
     ├─ √t Z : right marginals, wrong increments ⇒ not BM                  [–S12S13]│
     ├─ E[W_t]=0, E[W_t²]=t, Cov(W_s,W_t)=min(s,t) ; jointly normal         [S14]│
     └─ P(₂W<0 | ₁W>0), P(₁₂WW>0)                                          [S15]├─
 PROPERTIES [–S16S20]│
     ├─ Markov: E_s[W_t] = W_s ; GBM X_t = e^{W_t}                         [–S16S17]│
     └─ nowhere differentiable, fractal, recurrent ; reflection –W_t       [–S18S20]└─
 MARTINGALES [–S21S25]
      ├─ W_t ✓ ; W_t² – t ✓ ; W_t³ ✗                                        [–S21S23]
      └─ E[e^{W_t}] = e^{σ½²t} ; expo(W_t – σ½²t) exponential martingale     [–S24S25]
    ▼
LATER: S4 stochastic integrals and Itô (dW² = dt) ; S6 –BlackScholes ; S7 Girsanov
```

The storyline of the session.

- Session 2's arithmetic tree, with $\pm\sqrt{\Delta t}$ steps, is exactly a **scaled random walk**. Let the step size go to zero and you get **Brownian motion**, the continuous-time building block of every model in the course.
- The definition is about **increments**, not just marginals. That one point drives everything: it makes W Markov, gives the covariance $\min(s,t)$, and lets you compute any conditional expectation by writing $W_t = (W_t - W_s) + W_s$ (future independent piece + known piece).
- The **martingale** section is a toolkit for continuous-time pricing. The exponential martingale $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is exactly the discounted Black–Scholes stock under $\mathbb{Q}$; the $-\frac{1}{2}\sigma^2 t$ is the same convexity correction as in Sessions 1–2.

- Paths are continuous but **nowhere differentiable**, which is why Session 4 needs stochastic integrals and Itô's formula instead of ordinary calculus.

1 Concept entries

1.1 Symmetric random walk [S2]

The idea in words
The simplest model of "random up-and-down": start at 0 and flip a fair coin each step, +1 for heads, −1 for tails. With n steps there are 2^n equally likely paths. The position is not itself binomial; the *number of up-steps* is, and the position is an affine function of it. Mean 0 (fair coin), variance n (one per step): the variance grows linearly in the number of steps, the seed of "variance = time" for Brownian motion.

$$S_4 = 2B - 4, \qquad B \sim \text{Bin}(4, 0.5), \qquad \mathbb{P}(S_4 = 2k - 4) = \binom{4}{k}\left(\frac{1}{2}\right)^4.$$

where

S_4	position after 4 steps; reachable values $-4, -2, 0, 2, 4$; total paths $2^4 = 16$.
B	number of up steps among the 4.
$2B - 4$	B ups and $4 - B$ downs give $B - (4 - B)$.
$\binom{4}{k}\frac{1}{16}$	$\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$ for $S_4 = -4, \ldots, 4$.

Exam angle
△! [S2] writes $S_4 \sim \text{Bin}(4, 0.5)$. The number of up-steps is binomial; the position is $2B - 4$ (mean $2 \cdot 2 - 4 = 0$, variance $4 \cdot V[B] = 4 \cdot 1 = 4$).

1.2 History: from pollen to stock prices [S3–S8]

- **Robert Brown** (1827) observed the erratic motion of pollen particles in water and concluded it came from the particles themselves, not currents or evaporation [S3].
- **Einstein** showed the motion satisfies a clear statistical law [S4].
- **Louis Bachelier**, *Théorie de la spéculation* (1900), modelled stock-price fluctuations with Brownian motion [S5–S8]; it is still the building block of the reference continuous-time market model. We approach it as an infinitesimal random walk: smaller and smaller steps at ever more frequent intervals [S8].

1.3 From random walk to Brownian motion [S9–S10]

The idea in words
Squeeze the random walk: take n steps in time t, each of length t/n. To keep the total variance equal to t ("variance = elapsed time"), each jump must be $\pm\sqrt{t/n}$, the same $\sqrt{\Delta t}$ scaling as the Session 2 tree. As $n \to \infty$ the CLT turns the sum into a normal $\mathcal{N}(0, t)$, and because each step is a fresh coin, future moves never depend on the past. In the limit we get a continuous process with independent normal increments: Brownian motion.

Scaled random walk $W_n(t)$ [S9]: (1) $W_n(0) = 0$; (2) time spacing t/n; (3) up and down jumps equal, of size $\sqrt{t/n}$; (4) measure $\mathbb{P}$ with up/down probability $\frac{1}{2}$ everywhere.

$$W_n\left(\frac{i}{n}t\right) = W_n\left(\frac{i-1}{n}t\right) + \sqrt{\frac{t}{n}}\,X_i, \qquad 1 \le i \le n.$$

From the recursion to the limit [S9–S10] (the slide says "it can be shown")

$$W_n(t) = W_n(0) + \sum_{i=1}^{n} \sqrt{\tfrac{t}{n}} X_i = \sqrt{t} \cdot \frac{\sum_{i=1}^{n} X_i}{\sqrt{n}} \qquad \text{(telescope; } W_n(0) = 0\text{)}$$

$$\mathbb{E}\Big[\tfrac{1}{\sqrt{n}} \textstyle\sum X_i\Big] = 0, \quad V\Big[\tfrac{1}{\sqrt{n}} \textstyle\sum X_i\Big] = \tfrac{1}{n} \cdot n = 1 \qquad \text{(indep., } V[X_i] = 1\text{)}$$

$$\frac{\sum X_i}{\sqrt{n}} \xrightarrow{n\to\infty} Z \sim \mathcal{N}(0,1) \qquad \text{(CLT)}$$

$$W_n(t) \to \sqrt{t}\, Z \sim \mathcal{N}(0,t)$$

Why $\sqrt{t/n}$ and not t/n: the variance of the sum is $n \times (\text{jump})^2$. With jump $\sqrt{t/n}$ this is t; with t/n it would be $t^2/n \to 0$ (no randomness left).

Why the limit is Brownian motion [S10]:

- All marginal and conditional marginal distributions tend to normals.
- Future moves away from any position are independent of that position and of the whole history.
- A future displacement $W_n(s+t) - W_n(s)$ is (scaled) binomial with mean 0, variance t; by the CLT it tends to $\mathcal{N}(0,t)$.

1.4 Mathematical definition of Brownian motion [S11]

The idea in words

Wiener (1923) turned the limit into a definition. Four properties, and they are *necessary and sufficient*: start at zero; continuous paths; increments over non-overlapping intervals are independent and their distribution depends only on the interval length; and each increment is normal with variance equal to the elapsed time. Note that "$W_t \sim \mathcal{N}(0,t)$ for each t" is *not* on the list; it follows from (1) and (4), but on its own it is far too weak (next entry).

(1) $W_0 = 0$; (2) $t \mapsto W_t$ continuous;

(3) stationary, independent increments; (4) $W_t - W_s \sim \mathcal{N}(0, t-s)$, $0 \le s \le t$.

where

independent	$W_{t_2} - W_{t_1}$, $W_{t_3} - W_{t_2}, \ldots$ over non-overlapping intervals are independent; in particular $W_t - W_s$ is independent of the whole path up to s.
stationary	the distribution of $W_t - W_s$ depends only on $t - s$, not on s.
$\mathcal{N}(0, t-s)$	increment is normal, mean 0, variance = elapsed time.

Exam angle

How to test a candidate process X_t: check $X_0 = 0$; continuity; compute the distribution of $X_t - X_s$ (must be $\mathcal{N}(0, t-s)$); check it is independent of X_s (e.g. $\text{Cov}(X_t - X_s, X_s) = 0$ for jointly normal). △! [S11] spells the name "Weiner"; it is Wiener.

1.5 Example: $\sqrt{t}\,Z$ is not a Brownian motion [S12–S13]

The idea in words

$X_t = \sqrt{t}Z$ uses **one** normal draw for all times. Each snapshot is $\mathcal{N}(0,t)$, so marginals look right. But the whole path is a deterministic curve $\sqrt{t}$ scaled by one random number: once you see X_1 you know the entire future. Increments are therefore perfectly correlated with the past and have the wrong variance. Lesson: Brownian motion is defined by its increments.

Increment of $X_t = \sqrt{t}Z$ [S12]

$$\begin{aligned} X_{s+t} - X_s &= \sqrt{s+t}\,Z - \sqrt{s}\,Z = \left(\sqrt{s+t} - \sqrt{s}\right)Z && \text{(same } Z\text{)} \\ &\sim \mathcal{N}\left(0, \left(\sqrt{s+t} - \sqrt{s}\right)^2\right) && (V[cZ] = c^2) \\ &\sim \mathcal{N}\left(0,\ 2s + t - 2\sqrt{s(s+t)}\right) && \text{(expand)} \end{aligned}$$

Brownian motion needs variance t; here $2s + t - 2\sqrt{s(s+t)} < t$ for $s > 0$ (since $s < \sqrt{s(s+t)}$). And the increment $= (\sqrt{s+t} - \sqrt{s})Z$ is a multiple of $X_s = \sqrt{s}Z$: perfectly correlated, not independent. So X_t is not a Brownian motion.

[S13]: requiring $\mathcal{N}(0, t-s)$ increments independent of the past for every $s < t$ is *much stronger* than requiring $\mathcal{N}(0,t)$ marginals. Picture: W_t is rough and wanders; X_t is a smooth fan of $\sqrt{t}$ curves.

1.6 Statistical properties and joint normality [S14]

The idea in words

Everything follows from "increment $\sim \mathcal{N}(0, \text{elapsed time})$, independent of the past". The covariance is the clever one: W_s and W_t share the path up to s, and beyond that W_t only adds an independent piece. So they co-move through the common part only, and the covariance is the length of the shared time, $\min(s,t)$.

$$\mathbb{E}[W_t] = 0, \quad W_t \sim \mathcal{N}(0,t), \quad \mathbb{E}[W_t^2] = t, \quad \mathbb{E}[W_t - W_s] = 0,$$
$$V[W_t - W_s] = \mathbb{E}[(W_t - W_s)^2] = t - s, \qquad \mathrm{Cov}(W_s, W_t) = s \quad (s < t).$$

Covariance *[Beyond slides]* (slide states the result)

$$\begin{aligned} \mathrm{Cov}(W_s, W_t) &= \mathbb{E}[W_s W_t] - 0 \cdot 0 = \mathbb{E}\left[W_s\left((W_t - W_s) + W_s\right)\right] && \text{(add and subtract } W_s\text{)} \\ &= \mathbb{E}[W_s]\,\mathbb{E}[W_t - W_s] + \mathbb{E}[W_s^2] && \text{(independent increments)} \\ &= 0 + s = s = \min(s,t) && (\mathbb{E}[W_s^2] = s) \end{aligned}$$

Correlation $\rho(W_s, W_t) = \frac{s}{\sqrt{s}\sqrt{t}} = \sqrt{s/t}$.

For $t_1 < \cdots < t_n$, $(W_{t_1}, \ldots, W_{t_n})$ is **jointly normal**, mean 0, covariance $\Sigma_{ij} = \min(t_i, t_j)$:

$$\Sigma = \begin{pmatrix} t_1 & t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & t_2 & \cdots & t_2 \\ t_1 & t_2 & t_3 & \cdots & t_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & t_3 & \cdots & t_n \end{pmatrix}.$$

Why jointly normal *[Beyond slides]*: each W_{t_k} is a sum of the independent normal increments W_{t_1}, $W_{t_2} - W_{t_1}, \ldots$, and linear combinations of independent normals are jointly normal.

1.7 Probability examples [S15]

The idea in words

Any question about W at two times is solved by splitting the later value into "earlier value + independent increment". Here $W_2 = W_1 + Y$ with W_1, Y independent $\mathcal{N}(0,1)$. Two independent standard normals have a rotationally symmetric joint density, so the probability of a region bounded by lines through the origin is just its angle divided by 360°.

(a) $\mathbb{P}(W_2 < 0 \mid W_1 > 0)$ (worked here; the slide only poses it)

$$\begin{aligned}
\mathbb{P}(W_2 < 0 \mid W_1 > 0) &= \frac{\mathbb{P}(W_1 > 0,\ W_1 + Y < 0)}{\mathbb{P}(W_1 > 0)} && \text{(conditional probability)}\\
\mathbb{P}(W_1 > 0,\ Y < -W_1) &= \tfrac{45^\circ}{360^\circ} = \tfrac{1}{8} && \text{(wedge, see below)}\\
\mathbb{P}(W_2 < 0 \mid W_1 > 0) &= \frac{1/8}{1/2} = \frac{1}{4}.
\end{aligned}$$

The region $\{W_1 > 0,\ Y < -W_1\}$ in the (W_1, Y) plane is the wedge between the negative Y-axis and the ray $Y = -W_1$ ($W_1 > 0$): a 45° angle. The joint density of two independent standard normals is rotationally symmetric, so probability = angle/360°.

(b) $\mathbb{P}(W_1 W_2 > 0)$

Same sign means not (sign change). The two sign-change events $\{W_1 > 0, W_2 < 0\}$ and its mirror $\{W_1 < 0, W_2 > 0\}$ (reflection symmetry) each have probability $\frac{1}{8}$:

$$\mathbb{P}(W_1 W_2 > 0) = 1 - \tfrac{1}{8} - \tfrac{1}{8} = \tfrac{3}{4}.$$

General *[Beyond slides]*: $\mathbb{P}(W_s W_t > 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin \sqrt{s/t}$; for $s = 1$, $t = 2$: $\frac{1}{2} + \frac{1}{\pi} \cdot \frac{\pi}{4} = \frac{3}{4}$ ✓.

1.8 Markov property [S16]

The idea in words

A Markov process has no memory beyond its current value: knowing the whole history tells you nothing more about the future than knowing where you are now. Brownian motion is Markov because the future is "current value + an increment that is independent of the past". This is why option prices in Black–Scholes depend on the current S_t only, not on how it got there.

$$\mathbb{E}_t[X_T] = \mathbb{E}[X_T \mid X_t, X_{t-1}, X_{t-2}, \ldots, X_0] = \mathbb{E}[X_T \mid X_t].$$

Brownian motion is Markov [S16] ($0 < s < t$)

$$\begin{aligned}
\mathbb{E}_s[W_t] &= \mathbb{E}_s[W_t - W_s + W_s] && \text{(add and subtract } W_s\text{)}\\
&= \mathbb{E}_s[W_t - W_s] + \mathbb{E}_s[W_s] && \text{(linearity)}\\
&= 0 + W_s = W_s && \text{(increment independent, mean 0; } W_s \text{ known)}
\end{aligned}$$

Only W_s appears: to predict W_t you need only the present value. All independent-increment processes are Markov.

Exam angle

△! [S16] states the Markov property with conditional expectations; the property is about the whole conditional *distribution* (must hold for $\mathbb{E}[g(X_T) \mid \cdot]$ for every g). The expectation version is a consequence.

1.9 Geometric Brownian motion [S17]

The idea in words

Exponentiate a Brownian motion and you get a process that is always positive and moves in percentage terms: **geometric Brownian motion**, the Black–Scholes model of a stock. Expectations of $e^{\theta W_T}$ are just the normal MGF with variance T. Note $\mathbb{E}[e^{W_T}] = e^{T/2}$ grows even though $\mathbb{E}[W_T] = 0$: convexity again.

$$X_t = e^{W_t} \qquad \text{(full model } \textit{[Beyond slides]}\text{: } S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}\text{)}.$$

Slide example [S17] (worked here): $W_T \sim \mathcal{N}(0,T)$, $\mathbb{E}[e^{\theta W_T}] = e^{\frac{1}{2}\theta^2 T}$

$$\mathbb{E}[X_T] = \mathbb{E}[e^{1\cdot W_T}] = e^{\frac{1}{2}\cdot 1^2 \cdot T} = e^{T/2} \qquad \text{(MGF at } \theta = 1\text{)}$$
$$\mathbb{E}[X_T^2] = \mathbb{E}[e^{2W_T}] = e^{\frac{1}{2}\cdot 4 \cdot T} = e^{2T} \qquad (X_T^2 = e^{2W_T},\ \theta = 2)$$
$$V[X_T] = e^{2T} - e^{T} \qquad \textit{([Beyond slides])}$$

1.10 Path properties [S18–S20]

The idea in words
Zoom into a smooth curve and it looks like a straight line (that is what having a derivative means). Zoom into a Brownian path and it looks just as rough as before: it is self-similar. The reason is the $\sqrt{\Delta t}$ scaling: over a tiny interval h the move is of size $\sqrt{h}$, which is much larger than h, so the slope $\sqrt{h}/h = 1/\sqrt{h}$ blows up.

- **Nowhere differentiable [S18–S19]:** continuous everywhere, differentiable nowhere. Heuristic *[Beyond slides]*: $\frac{W_{t+h}-W_t}{h} \sim \mathcal{N}(0, 1/h)$, whose variance explodes as $h \to 0$.
- **Self-similar / fractal [S18–S19]:** looks the same at every scale. *[Beyond slides]*: $\frac{1}{\sqrt{c}}W_{ct}$ is again a Brownian motion for any $c > 0$.
- **Recurrence [S19]:** hits every real value eventually (w.p. 1), returns to zero however far it wanders, and once it hits a value it hits it again infinitely often.
- **Reflection [S20]:** $W_t \mapsto -W_t$ is again a Brownian motion, because each increment's distribution $\mathcal{N}(0, t-s)$ is symmetric about 0.

Why scaling and reflection preserve Brownian motion *[Beyond slides]* (check the four properties)
For $B_t = \frac{1}{\sqrt{c}}W_{ct}$: $B_0 = 0$; continuous; increments $B_t - B_s = \frac{1}{\sqrt{c}}(W_{ct} - W_{cs})$ are built from non-overlapping increments of W, hence independent; and $V[B_t - B_s] = \frac{1}{c} \cdot c(t-s) = t - s$. For $-W_t$: same, with $V[-(W_t - W_s)] = t - s$ and symmetry of the normal.

1.11 Brownian martingales [S21–S23]

The idea in words
A martingale is a "fair game": given everything known today, the expected future value is today's value. To check, always use the same trick: split $W_t = (W_t - W_s) + W_s$, expand, then use two facts: the increment is independent of the past (so its conditional moments are ordinary moments: mean 0, second moment $t - s$, third moment 0), and W_s is known (so it comes out). If the result contains anything besides the original expression at time s, the leftover is a drift and you can often subtract it to make a martingale (a "compensator").

Definition [S21]: $M_t = \mathbb{E}_t[M_T]$ for all $T \geq t$ *([Beyond slides]*: also $\mathbb{E}|M_t| < \infty$ and M_t known at t).

W_t is a martingale [S21]
$\mathbb{E}_s[W_t] = \mathbb{E}_s[W_t - W_s] + \mathbb{E}_s[W_s] = 0 + W_s = W_s$ (same computation as Markov).

$W_t^2 - t$ is a martingale [S22] (the slide skips the middle steps)

$$\begin{aligned}
\mathbb{E}_s[W_t^2 - t] &= \mathbb{E}_s\big[(W_t - W_s + W_s)^2\big] - t && \text{(split; } t \text{ is not random)} \\
&= \mathbb{E}_s\big[(W_t - W_s)^2 + 2W_s(W_t - W_s) + W_s^2\big] - t && \text{(expand)} \\
&= \underbrace{\mathbb{E}[(W_t - W_s)^2]}_{t-s} + 2W_s \underbrace{\mathbb{E}[W_t - W_s]}_{0} + W_s^2 - t && \text{(independence; } W_s \text{ known)} \\
&= W_s^2 - s && (t - s - t = -s)
\end{aligned}$$

Without the $-t$, $\mathbb{E}_s[W_t^2] = W_s^2 + (t-s)$: W_t^2 drifts up at rate 1. Subtracting t removes exactly that drift.

W_t^3 is not a martingale [S23]

$$\begin{aligned}
\mathbb{E}_s[W_t^3] &= \mathbb{E}_s\big[(W_t - W_s + W_s)^3\big] && \text{(split)} \\
&= \mathbb{E}_s\big[(W_t - W_s)^3 + 3(W_t - W_s)^2 W_s + 3(W_t - W_s)W_s^2 + W_s^3\big] && \text{(binomial expansion)} \\
&= \underbrace{\mathbb{E}[(W_t - W_s)^3]}_{0} + 3W_s \underbrace{\mathbb{E}[(W_t - W_s)^2]}_{t-s} + 3W_s^2 \underbrace{\mathbb{E}[W_t - W_s]}_{0} + W_s^3 && \text{(odd moments of } \mathcal{N}(0,\cdot) \text{ vanish)} \\
&= W_s^3 + 3(t-s)W_s \;\neq\; W_s^3
\end{aligned}$$

[Beyond slides] Compensator: $\mathbb{E}_s[3tW_t] = 3tW_s$, so $\mathbb{E}_s[W_t^3 - 3tW_t] = W_s^3 + 3(t-s)W_s - 3tW_s = W_s^3 - 3sW_s$: $W_t^3 - 3tW_t$ is a martingale.

Exam angle

△! [S23] concludes that W_t^3 "doesn't satisfy the definition of a standard Brownian motion"; it should say *of a martingale*. Exam pattern: "show $f(W_t, t)$ is (not) a martingale" ⇒ split, expand, use moments of $\mathcal{N}(0, t-s)$: $0,\ t-s,\ 0,\ 3(t-s)^2$.

1.12 Exponential martingale [S24–S25]

The idea in words

$e^{\sigma W_t}$ grows on average (convexity), at the rate $e^{\frac{1}{2}\sigma^2 t}$ given by the MGF. Divide by exactly that growth and what is left is a martingale: $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$. This is the continuous-time version of "discounted prices are $\mathbb{Q}$-martingales": the discounted Black–Scholes stock is $e^{-rt}S_t = S_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$. It is also the Radon–Nikodym density used in Girsanov's theorem (Session 7).

MGF recap [S24]: $X \sim \mathcal{N}(\mu, \sigma^2)$: $\mathbb{E}[e^{\theta X}] = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}$; $Z \sim \mathcal{N}(0,1)$: $\mathbb{E}[e^{\theta Z}] = e^{\frac{1}{2}\theta^2}$.

Slide example [S24] (worked here): $\mathbb{E}[e^{\sigma W_t}]$

$W_t \sim \mathcal{N}(0, t)$, so use the MGF with $\mu = 0$, variance t, $\theta = \sigma$: $\mathbb{E}[e^{\sigma W_t}] = e^{0 + \frac{1}{2}t\sigma^2} = e^{\frac{1}{2}\sigma^2 t}$. (Equivalently $W_t = \sqrt{t}Z$, $\mathbb{E}[e^{\sigma\sqrt{t}Z}] = e^{\frac{1}{2}\sigma^2 t}$.)

$\exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ is a martingale [S25]

$$\begin{aligned}
\mathbb{E}_s\big[e^{\sigma W_t - \frac{1}{2}\sigma^2 t}\big] &= \mathbb{E}_s\big[e^{\sigma(W_t - W_s)} \cdot e^{\sigma W_s - \frac{1}{2}\sigma^2 t}\big] && (W_t = (W_t - W_s) + W_s) \\
&= e^{\sigma W_s - \frac{1}{2}\sigma^2 t} \cdot \mathbb{E}_s\big[e^{\sigma(W_t - W_s)}\big] && \text{(known factor out)} \\
&= e^{\sigma W_s - \frac{1}{2}\sigma^2 t} \cdot e^{\frac{1}{2}\sigma^2(t-s)} && \text{(MGF of } \mathcal{N}(0, t-s)) \\
&= e^{\sigma W_s - \frac{1}{2}\sigma^2 s} && \text{(exponents: } -\tfrac{1}{2}\sigma^2 t + \tfrac{1}{2}\sigma^2(t-s))
\end{aligned}$$

where

$e^{\sigma(W_t - W_s)}$	the future part: depends only on the increment, independent of the past.
$-\frac{1}{2}\sigma^2 t$	compensator: exactly cancels the convexity growth $e^{\frac{1}{2}\sigma^2 t}$, leaving no drift.

1.13 Implementation *[Beyond slides]* (tested)

```
import numpy as np
rng = np.random.default_rng(0)
n = 2_000_000
W1 = rng.standard_normal(n)
W2 = W1 + rng.standard_normal(n)              # W2 - W1 ~ N(0,1), independent of W1
print(round(np.mean(W2[W1 > 0] < 0), 3))      # P(W2<0 | W1>0) = 0.25
print(round(np.mean(W1*W2 > 0), 3))           # P(W1 W2 > 0)   = 0.75
T = 1.0
WT = np.sqrt(T) * rng.standard_normal(n)
print(round(np.mean(np.exp(WT)), 3), round(np.exp(T/2), 3))      # 1.649 1.649
print(round(np.mean(np.exp(2*WT)), 2), round(np.exp(2*T), 2))    # 7.41 7.39 (MC noise)
```

2 Comparison tables

2.1 Brownian motion vs $\sqrt{t}\,Z$ [S12–S13]

	Brownian motion W_t	$X_t = \sqrt{t}Z$
Marginal at t	$\mathcal{N}(0,t)$	$\mathcal{N}(0,t)$
Increment over $[s, s+t]$	$\mathcal{N}(0,t)$	$\mathcal{N}(0,(\sqrt{s+t}-\sqrt{s})^2)$
Increment vs past	independent	perfectly correlated
Paths	rough, nowhere differentiable	smooth curves
Brownian motion?	yes	no

2.2 Random walk vs Brownian motion

	Scaled random walk $W_n(t)$	Brownian motion W_t
Time	discrete, step t/n	continuous
Step	$\pm\sqrt{t/n}$, probability $\frac{1}{2}$	$\mathrm{d}W_t$, infinitesimal
Distribution at t	scaled binomial, mean 0, var t	$\mathcal{N}(0,t)$ exactly
Increments	independent	independent, stationary, normal
Link	$W_n(t) \to W_t$ as $n \to \infty$ (CLT)	

2.3 Martingale or not?

Process	$\mathbb{E}_s[\cdot]$	Martingale?
W_t	W_s	yes [S21]
W_t^2	$W_s^2 + (t-s)$	no (drift $+1$)
$W_t^2 - t$	$W_s^2 - s$	yes [S22]
W_t^3	$W_s^3 + 3(t-s)W_s$	no [S23]
$W_t^3 - 3tW_t$	$W_s^3 - 3sW_s$	yes *[Beyond slides]*
$e^{\sigma W_t}$	$e^{\sigma W_s + \frac{1}{2}\sigma^2(t-s)}$	no
$e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$	$e^{\sigma W_s - \frac{1}{2}\sigma^2 s}$	yes [S25]

2.4 Path properties

Continuity	continuous everywhere [S11, S19]
Differentiability	nowhere differentiable [S18–S19]
Scale	self-similar, fractal [S19]
Recurrence	hits every level; returns to 0 infinitely often [S19]
Invariance	reflection $-W_t$ [S20]; scaling $c^{-1/2}W_{ct}$ *[Beyond slides]*

3 Master formula sheet

Random walk and limit [S2, S9–S10]: $W_n(\frac{i}{n}t) = W_n(\frac{i-1}{n}t) + \sqrt{t/n}X_i$; $W_n(t) = \sqrt{t}\frac{\sum X_i}{\sqrt{n}} \to \mathcal{N}(0,t)$.

Definition and moments [S11–S14]

- $W_0 = 0$; continuous; independent stationary increments; $W_t - W_s \sim \mathcal{N}(0, t-s)$.
- $\mathbb{E}[W_t] = 0$, $\mathbb{E}[W_t^2] = t$, $V[W_t - W_s] = t - s$, $\mathrm{Cov}(W_s, W_t) = \min(s,t)$, $\rho = \sqrt{s/t}$.

Probabilities [S15]: $\mathbb{P}(W_2 < 0 \mid W_1 > 0) = \frac{1}{4}$; $\mathbb{P}(W_1 W_2 > 0) = \frac{3}{4}$; $\mathbb{P}(W_s W_t > 0) = \frac{1}{2} + \frac{1}{\pi} \arcsin \sqrt{s/t}$ *[Beyond slides].*

Markov and martingales [S16, S21–S25]

- Markov: $\mathbb{E}_t[X_T] = \mathbb{E}[X_T \mid X_t]$; $\mathbb{E}_s[W_t] = W_s$.
- Martingale: $M_t = \mathbb{E}_t[M_T]$. Martingales: W_t, $W_t^2 - t$, $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$. Not: W_t^3.

Exponentials [S17, S24]: $\mathbb{E}[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}$; $\mathbb{E}[e^{W_T}] = e^{T/2}$, $\mathbb{E}[e^{2W_T}] = e^{2T}$.

4 Glossary

- **Brownian motion / Wiener process:** continuous process, $W_0 = 0$, independent $\mathcal{N}(0, t-s)$ increments [S11].
- **Exponential martingale:** $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ [S25]. **Fractal / self-similar:** statistically the same at every scale [S19].
- **Geometric Brownian motion:** exponential of a Brownian motion; Black–Scholes price model [S17].
- **Independent / stationary increments:** increments over disjoint intervals independent / distribution depends only on interval length [S11].
- **Jointly normal:** any vector of W_{t_i} is multivariate normal [S14].
- **Markov property:** the future depends on the past only through the present [S16]. **Martingale:** $M_t = \mathbb{E}_t[M_T]$ [S21].
- **Recurrence:** returns to every level infinitely often [S19]. **Reflection invariance:** $-W_t$ is a Brownian motion [S20].
- **Scaled random walk:** steps $\pm\sqrt{t/n}$ every t/n [S9].

5 Self-test

Questions

Q1. ([S2]) For a 4-step symmetric random walk, give the distribution, mean and variance of S_4.

Q2. ([S9–S10]) Why are the scaled walk's jumps $\pm\sqrt{t/n}$ and not $\pm t/n$?

Q3. ([S11]) State the four defining properties of Brownian motion.

Q4. ([S12]) Is $X_t = \sqrt{t}Z$ a Brownian motion? Is $-W_t$? Is $\frac{1}{\sqrt{c}} W_{ct}$?

Q5. ([S14]) Show $\mathrm{Cov}(W_s, W_t) = s$ for $s < t$ and write the covariance matrix of (W_1, W_2, W_3).

Q6. ([S15]) Find $\mathbb{P}(W_2 < 0 \mid W_1 > 0)$ and $\mathbb{P}(W_1 W_2 > 0)$.

Q7. ([S17]) For $X_t = e^{W_t}$ find $\mathbb{E}[X_T]$, $\mathbb{E}[X_T^2]$, $V[X_T]$.

Q8. ([S21–S23]) Show W_t and $W_t^2 - t$ are martingales and W_t^3 is not. Find the compensated martingale built from W_t^3.

Q9. ([S24–S25]) Find $\mathbb{E}[e^{\sigma W_t}]$ and show $e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ is a martingale.

Q10. ([S16]) What is the Markov property and why does Brownian motion have it?

Solutions

S1. $S_4 = 2B - 4$, $B \sim \mathrm{Bin}(4, \frac{1}{2})$: $\mathbb{P}(S_4 = -4, \ldots, 4) = \frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$. Mean 0, variance 4.

S2. Variance of the sum is $n \times (\text{jump})^2$; with $\sqrt{t/n}$ this is t. With t/n it would be $t^2/n \to 0$.

S3. $W_0 = 0$; continuous paths; stationary, independent increments; $W_t - W_s \sim \mathcal{N}(0, t-s)$.

S4. $\sqrt{t}Z$: no (wrong increment variance, increments not independent). $-W_t$: yes. $c^{-1/2}W_{ct}$: yes; increments $\mathcal{N}(0, \frac{c(t-s)}{c})$, independent.

S5. $\mathbb{E}[W_s W_t] = \mathbb{E}[W_s(W_t - W_s)] + \mathbb{E}[W_s^2] = 0 + s$. $\Sigma = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$.

S6. $W_2 = W_1 + Y$; region $\{W_1 > 0, Y < -W_1\}$ is a 45° wedge: $\frac{1}{8}$. Conditional $\frac{1/8}{1/2} = \frac{1}{4}$. Same sign $1 - 2 \cdot \frac{1}{8} = \frac{3}{4}$.

S7. $e^{T/2}$, e^{2T}, $e^{2T} - e^{T}$.

S8. See 1.11. $\mathbb{E}_s[W_t^3] = W_s^3 + 3(t-s)W_s$; $\mathbb{E}_s[3tW_t] = 3tW_s$, so $W_t^3 - 3tW_t$ is a martingale.

S9. $e^{\frac{1}{2}\sigma^2 t}$. Split off the increment, pull out the known factor, MGF of $W_t - W_s$: get $e^{\sigma W_s - \frac{1}{2}\sigma^2 s}$.

S10. The conditional distribution of the future depends only on the present value. $W_t = W_s + (W_t - W_s)$ and the increment is independent of the whole past, so only W_s matters.

Appendix A. Slide coverage checklist

Slide	Content	Section
S2	4-step symmetric random walk	1.1
S3–S8	Brown, Einstein, Bachelier; BM as infinitesimal random walk	1.2
S9–S10	Scaled random walk; CLT limit; independent increments	1.3
S11	Mathematical definition	1.4
S12–S13	$\sqrt{t}Z$ is not BM; marginals vs increments	1.5
S14	Moments, covariance, joint normality	1.6
S15	Probability examples	1.7
S16	Markov property	1.8
S17	Geometric Brownian motion; example	1.9
S18–S20	Differentiability, fractal, recurrence, reflection	1.10
S21–S23	Martingale definition; W_t, $W_t^2 - t$, W_t^3	1.11
S24–S25	MGF; $\mathbb{E}[e^{\sigma W_t}]$; exponential martingale	1.12

Appendix B. Flag summary (△!)

1. [S2] $S_4 \sim \text{Bin}(4, 0.5)$: the number of ups is binomial; $S_4 = 2B - 4$.
2. [S11] "Weiner" should be Wiener.
3. [S16] The Markov property concerns conditional distributions; the expectation form is a consequence.
4. [S21] The martingale definition also needs integrability and adaptedness (standard, omitted).
5. [S23] "doesn't satisfy the definition of a standard Brownian motion" should read "of a martingale".
6. [S15, S17, S24] Examples posed without worked answers; the solutions here are my own.